

From your previous **Assignment 1 in SU1**, you created your example applying OOP. Congratulations! You managed to achieve the first part of your task. As an aspiring young Computer Scientist, you will often come across Data Structures and Algorithms in your academics and career, and in most cases be told to think outside the box. Let's put that to the test and see how well we can think! Shall We!!! It is mandatory to use the same .java files from **Assignment 1** to work on this assignment.

From **Assignment 1** you created one superclass and two subclasses, for **Assignment 2**, you will use the same superclass and subclasses. Note that all classes must have *variables, constructors, accessors, mutators, toString and abstract methods*. Your example must have a test program that use the created classes and demonstrates polymorphism and inheritance, as well as the implicit and explicit calling of the *toString method*. The program will randomly create instances of these subclasses, store them in an array, and then sort the array.

For **ASSIGNMENT 2**, you must create an automated Time Complexity Analyzer (ComplexityAnalyzer.java) that reads your Java source files **programmatically and computes** time complexity for all the classes using all three approaches studied in SU2 namely, **tau-notation, simplified method and Big-Oh notation**. **You must NOT hardcode or manually write the time complexities. The program must analyze the code dynamically.**

What the Analyzer Must Do

- Read each .java file
- Parse methods from the source code
- Programmatically detect the:
 - assignments (=)
 - Arithmetic operations (+, -, *, /)
 - Comparisons (==, !=, <, >, <=, >=)
 - Loops (for, while)
 - Method calls
 - String concatenations
- Compute complexity for each method using all three approaches for all the classes
- Output a formatted report for all classes

Important rules for all assignments

- **Only work submitted on e-fundi will be marked**
- **No work may be submitted on any other platform other than e-fundi.**
- **Only working code will be marked.**
- **Code will be checked for plagiarism.**
- **Remember this is not a group assignment. Enjoy!!!**